Universidad de Concepción

FACULTAD DE CIENCIAS BIOLÓGICAS
Universidad de Concepción

Quality Control

30 AÑOS FCB UDEC

Formato FASTQ

Is an extension of the FASTA format carrying quality values associated with each base

Start symbol

Sequence ID

Sequence

```
@read00179
AGTCTGATATGCTGTACCTATTATAATTCTAGGCGCTCAT
GCCCGCGGATATCGTAGCTATATGCTTCA
+
8;ACCCD?DD???@B9<9<CAC@=AAAA8A;B<A@882,+
495;;3990,02..-&-&-*,,,,(0**#
```

Separator line

Encoded quality values,
one symbol per nucleotide

https://www.ncbi.nlm.nih.gov/sra/docs/submitformats/#fastq-files

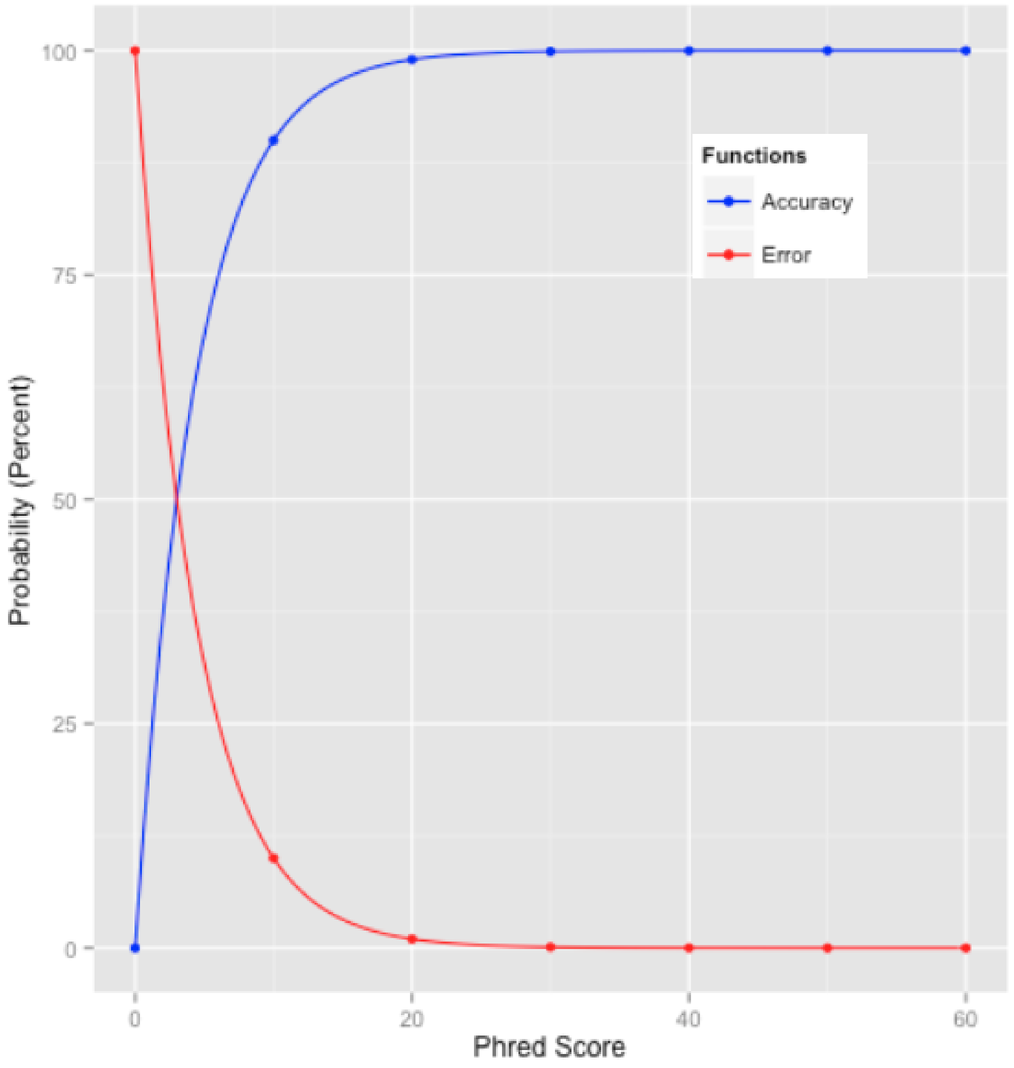
Quality Value are expressed by phred score

Sequencing quality scores measure the probability that a base is called incorrectly. With sequencing by synthesis (SBS) technology, each base in a read is assigned with a quality score by a phred algorithm.

The sequencing quality score of a given base, Q, is defined by the following equation:

$$Q = -10\log10(e)$$

where e is the estimated probability of the base call being wrong.



How quality value are generated

Illumina quality scores are calculated for each base call in a two-step process:

1. Quality predictor values are observable properties of clusters from which base calls are extracted.
    a) intensity profiles
    b) signal-to-noise ratios

2. A quality model, also known as a quality table or Q-table, lists combinations of quality predictor values and relates them to corresponding quality scores.

Table 1: Q-Scores and Error Probabilities

Quality Score	Error Probability
Q40	0.0001 (1 in 10,000)
Q30	0.001 (1 in 1,000)
Q20	0.01 (1 in 100)
Q10	0.1 (1 in 10)

Quality value code

The quality values are coded on letters and symbols

The coding system is the ASCII (American Standard Code for Information Interchange)

```
SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX......................
...............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII......................
.................................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.....................
LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL....................................................
!"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
|                         |    |        |                              |                     |
33                        59   64       73                            104                  126
0........................26...31.......40
                         -5....0........9..............................40
                               0........9..............................40
                                  3.....9...............................41
0.2.......................26...31........41

S - Sanger        Phred+33,  raw reads typically (0, 40)
X - Solexa        Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64,  raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64,  raw reads typically (3, 41)
    with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
    (Note: See discussion above).
L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

```
!"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJ
|         |         |         |         |
Q0        Q10       Q20       Q30       Q40
bad       maybe     ok        good      excellent
```

Quality value exercise

```
@SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=72
GGGTGATGGCCGCTGCCGATGGCGTCAAATCCCACCAAGTTACCCTTAACAACTTAAGGGTTTTCAAATAGA
+SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=72
IIIIIIIIIIIIIIIIIIIIIIIIIIIIII9IG9ICIIIIIIIIIIIIIIIIIIIIDIIIIIII>IIIIII/
```

```
 SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
 ..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX......................
 ...............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII......................
 .................................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.....................
 LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL....................................................
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
 |                         |    |        |                              |                     |
33                        59   64       73                            104                   126
 0........................26...31.......40
                          -5....0........9.............................40
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```
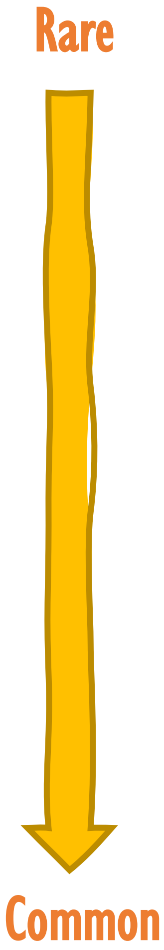

Additional quality issues on sequencing data

The sequences might contain errors such as:

- "Trash" Reads (Uncharacterized errors)
- Duplicated Reads (Low complexity libraries and PCR duplicares)
- Reading into the adapters (short fragments in comparison with read length)
- Error Indel (Deletion or insertion of a base during sequencing)
- Undetermined Base (base calling base was uncertain and it is replaced by an N)
- Substitution errors (wrong base calling base)



How data quality impacts on my results?

The GIGO (GARBAGE IN – GARBAGE OUT) PARADIGM

To avoid starting with bad quality data, we perform a quality control analysis

Quality analysis and global statistics

Due to different source errors that might be present in our sequences, it is crucial to perform a quality check **BEFORE** we start the analysis.

The gold standard program to do quality checks is **FASTQC**

Babraham Institute

Babraham Bioinformatics

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FastQC

Function	A quality control tool for high throughput sequence data.
Language	Java
Requirements	A suitable Java Runtime Environment The Picard BAM/SAM Libraries (included in download)
Code Maturity	Stable. Mature code, but feedback is appreciated.
Code Released	Yes, under GPL v3 or later.
Initial Contact	Simon Andrews

Download Now

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

FASTQC Quality analysis

- Global statistics (Number of sequences, read length, %GC)
- Phred quality score (Per base and global Q-score)
- %GC (most important for genome assembly)
- Undetermined Base (% N bases along the reads)
- Phred quality score (Per base and global Q-score)
- Distribution of read length
- Read Duplication
- Overrepresented sequences
- Adapter Content

FASTQC Report

Good Quality Run

Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

OK

warning

Fail

Bad Quality Run

Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
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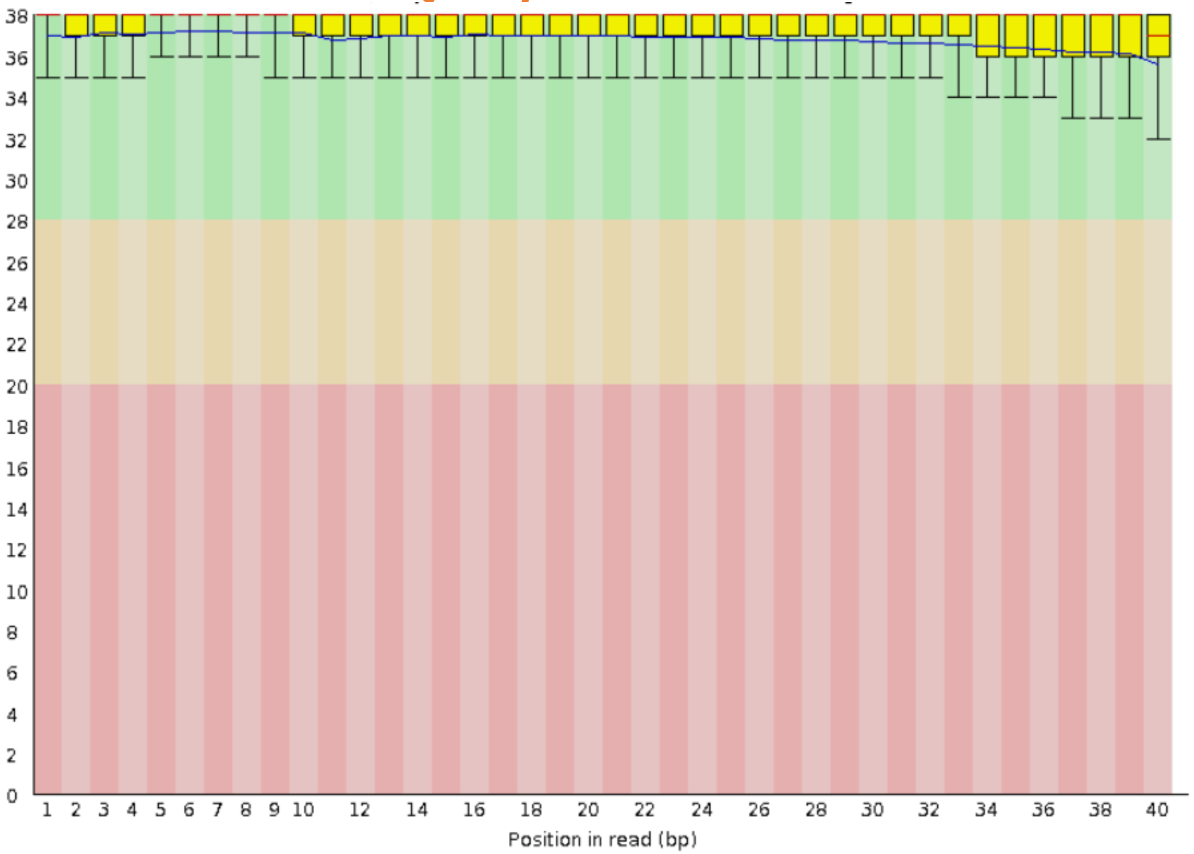
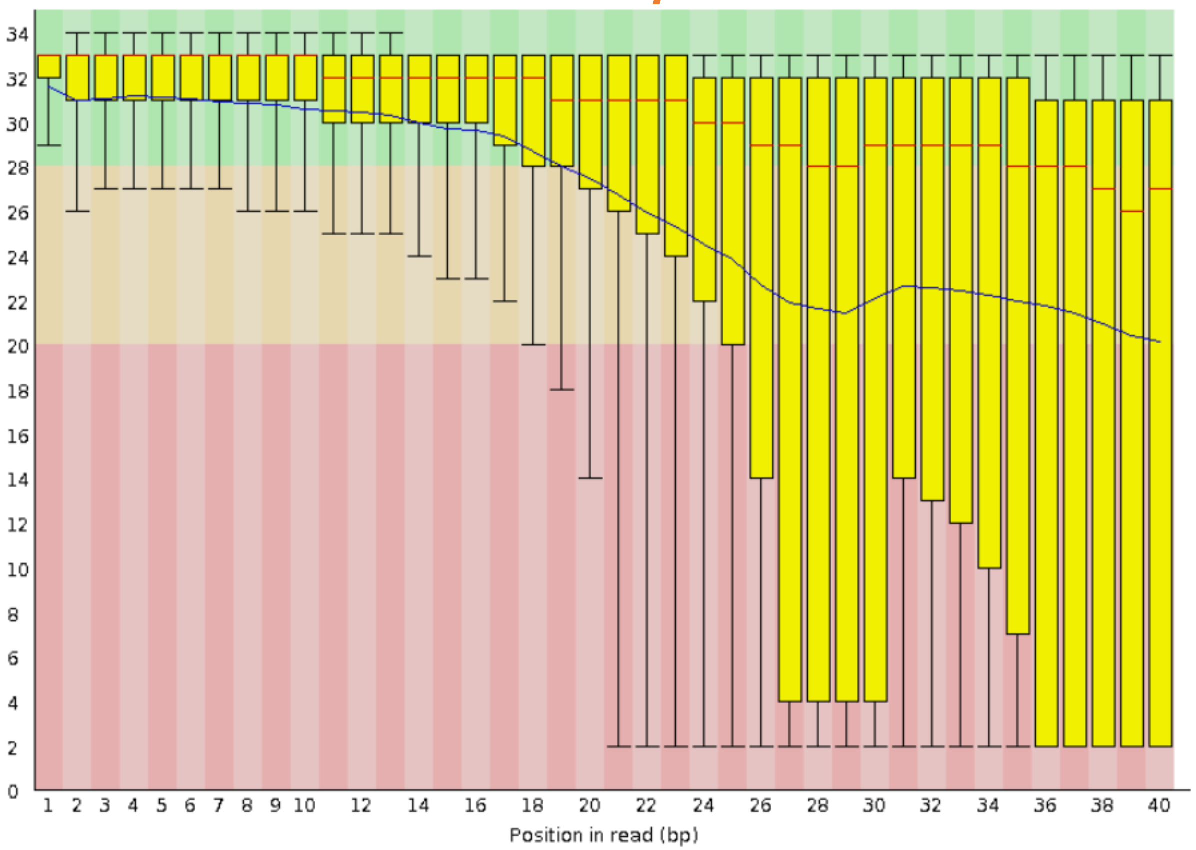
Per Base quality

Good Quality Run



Bad Quality Run



median value

mean value

inter-quartile range (25-75%)

whiskers 10% and 90% of points

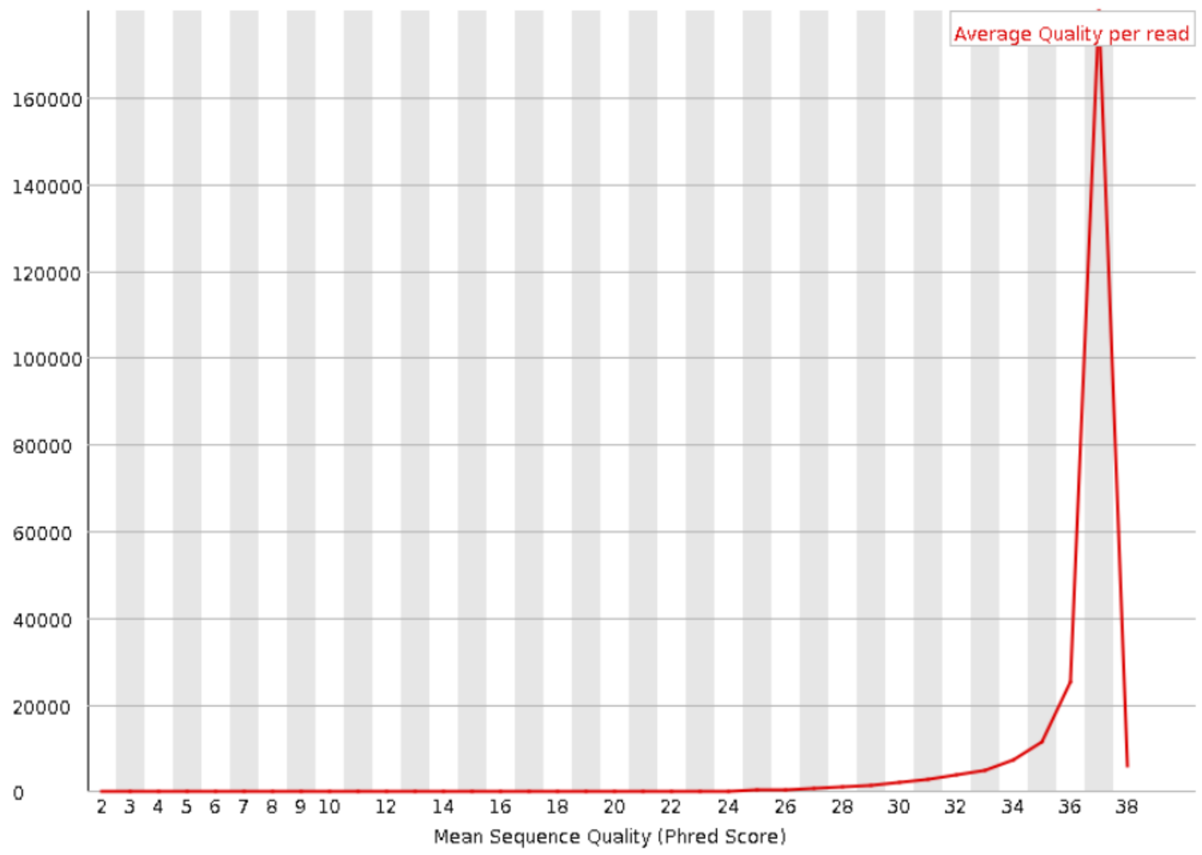
How can I Improve?:

Trim bad quality bases from 3' (<20-25 Phred score)

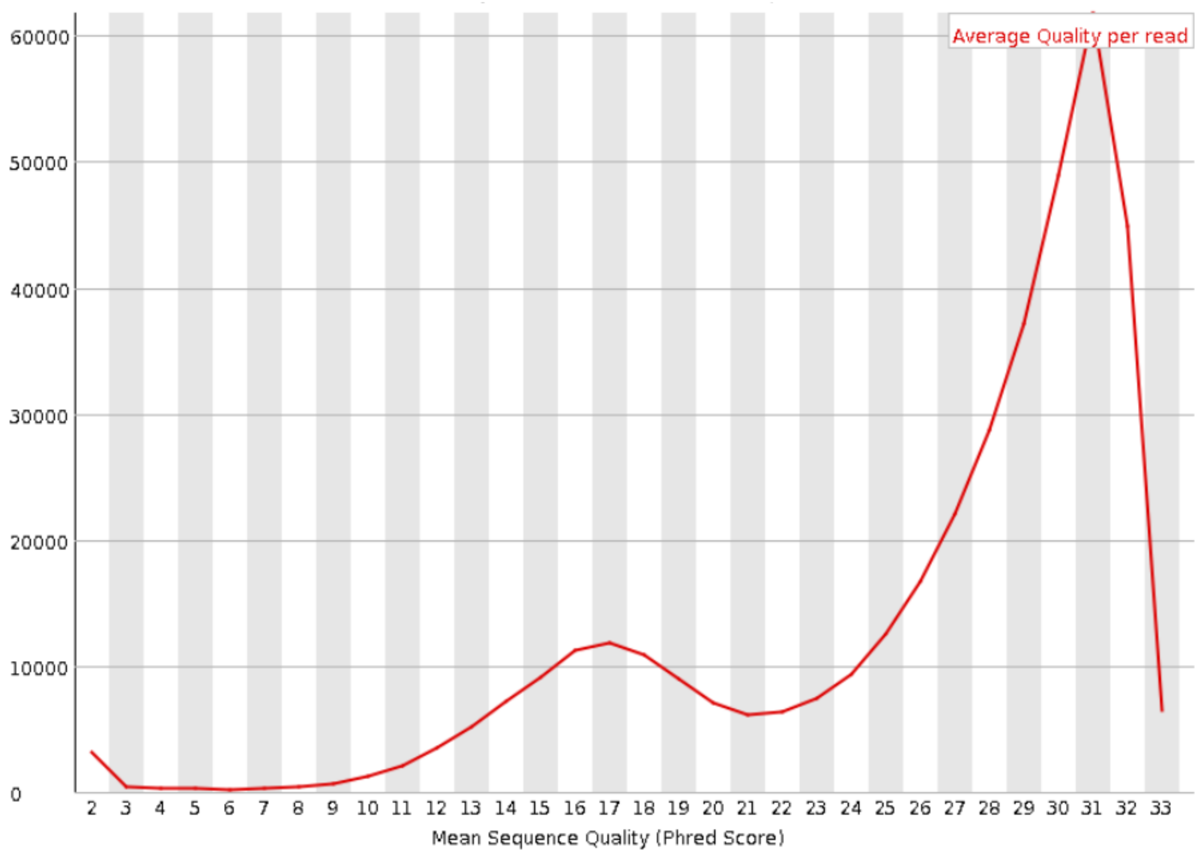
Do not keep reads shorter that 35bp

Per sequence quality value

Good Quality Run



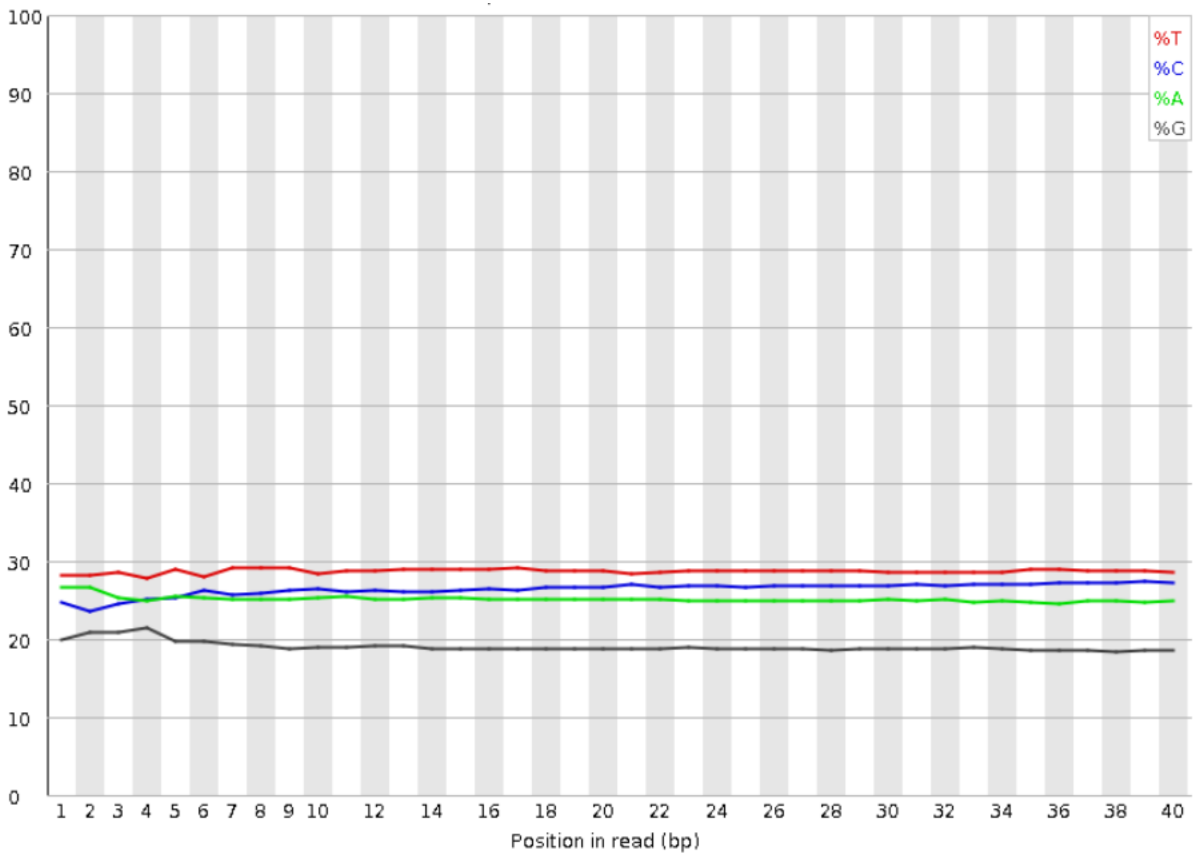
Bad Quality Run



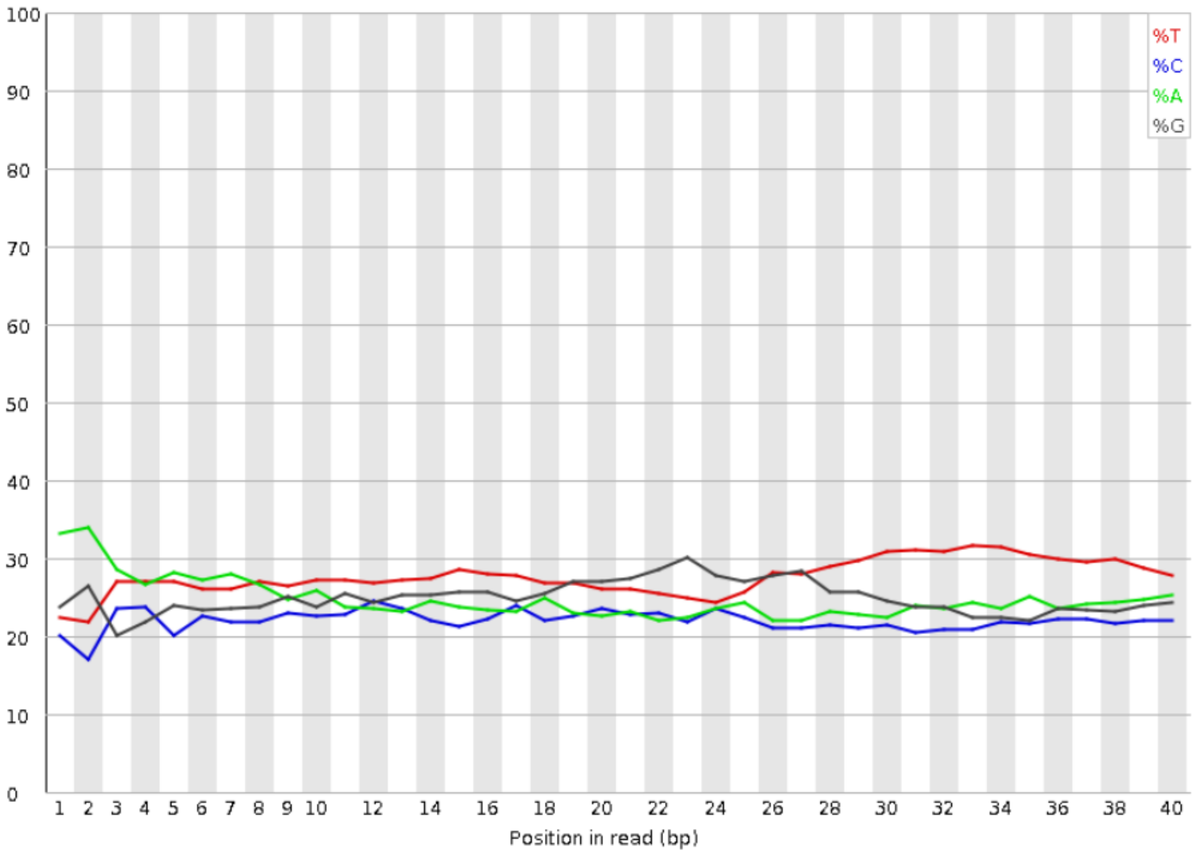
How can I Improve?:
Remove reads with average phred score smaller than 20

Base content

Good Quality Run



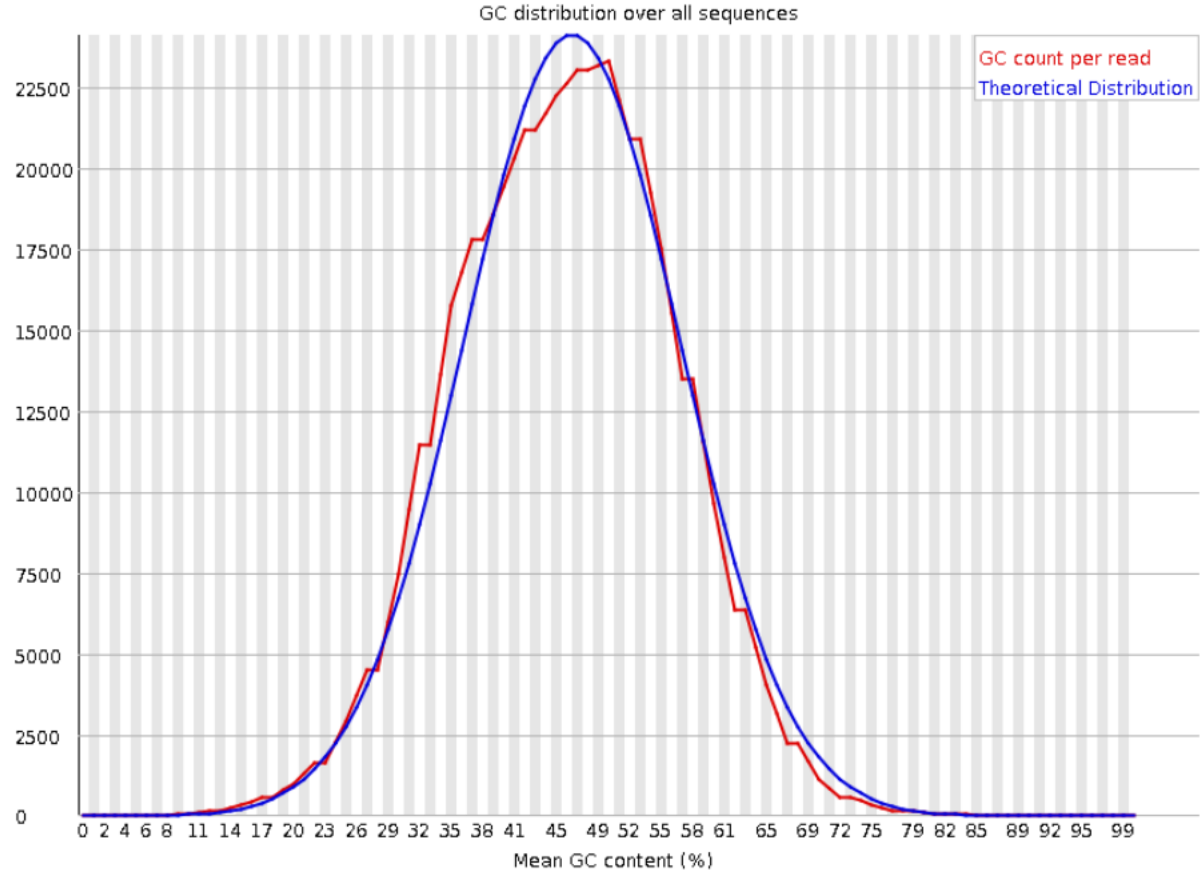
Bad Quality Run



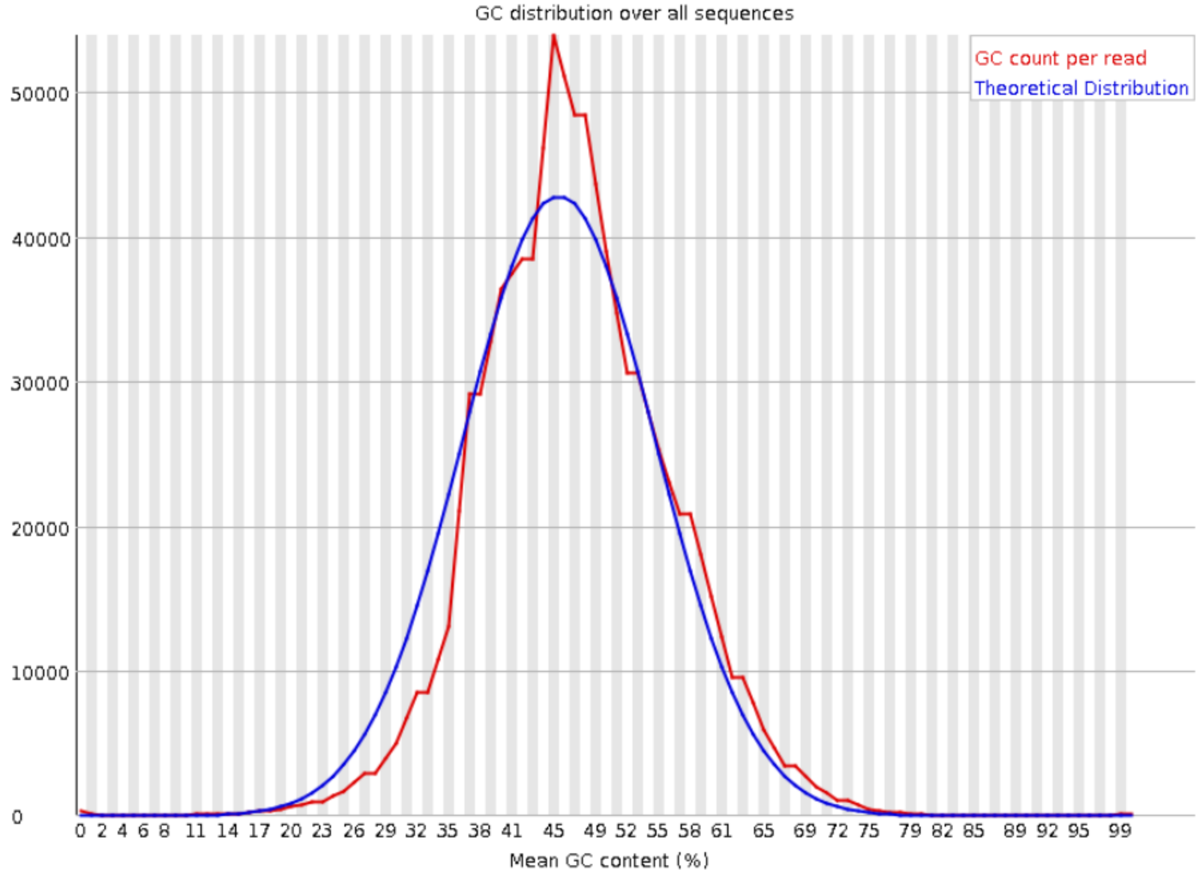
How can I Improve?:
If base bias is present at the ends of the read, we can cut bases from the end and start of the read.

GC content

Good Quality Run



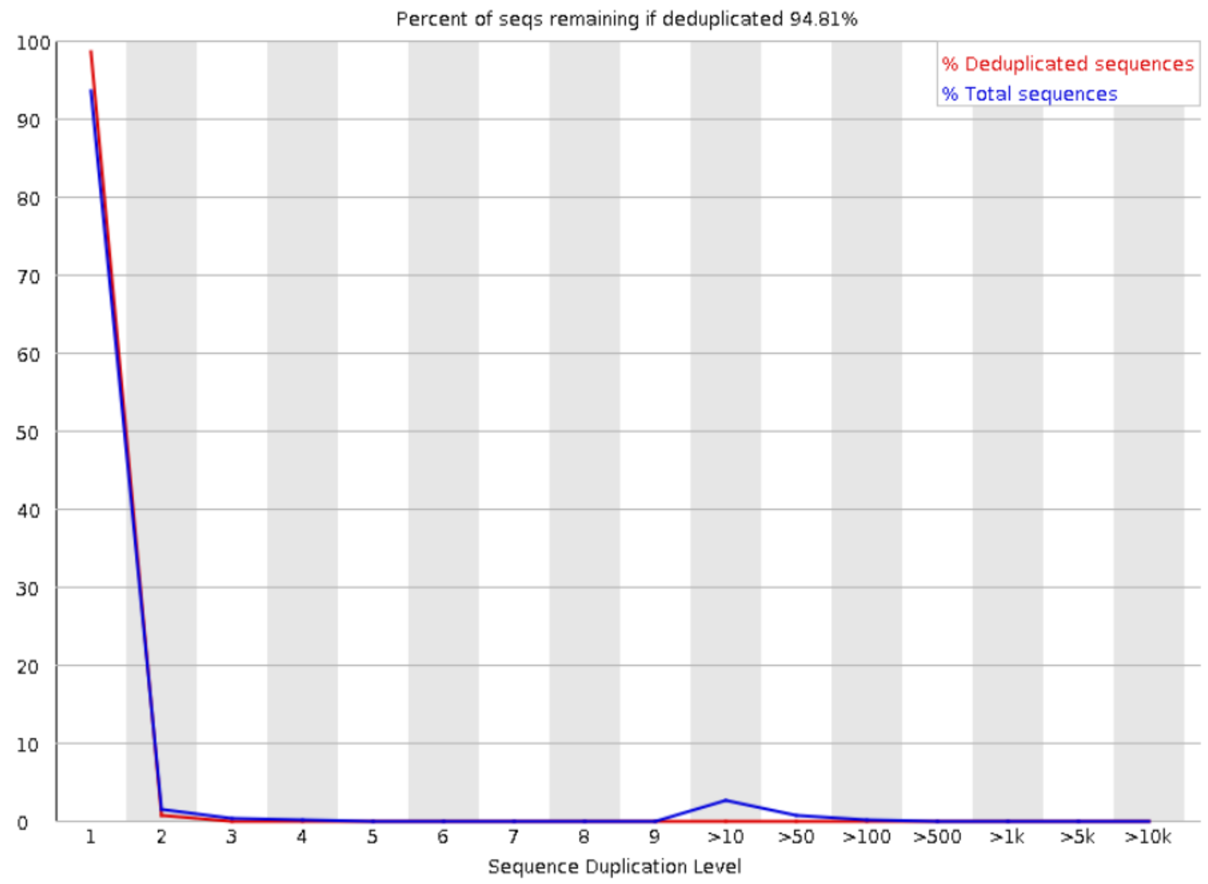
Bad Quality Run



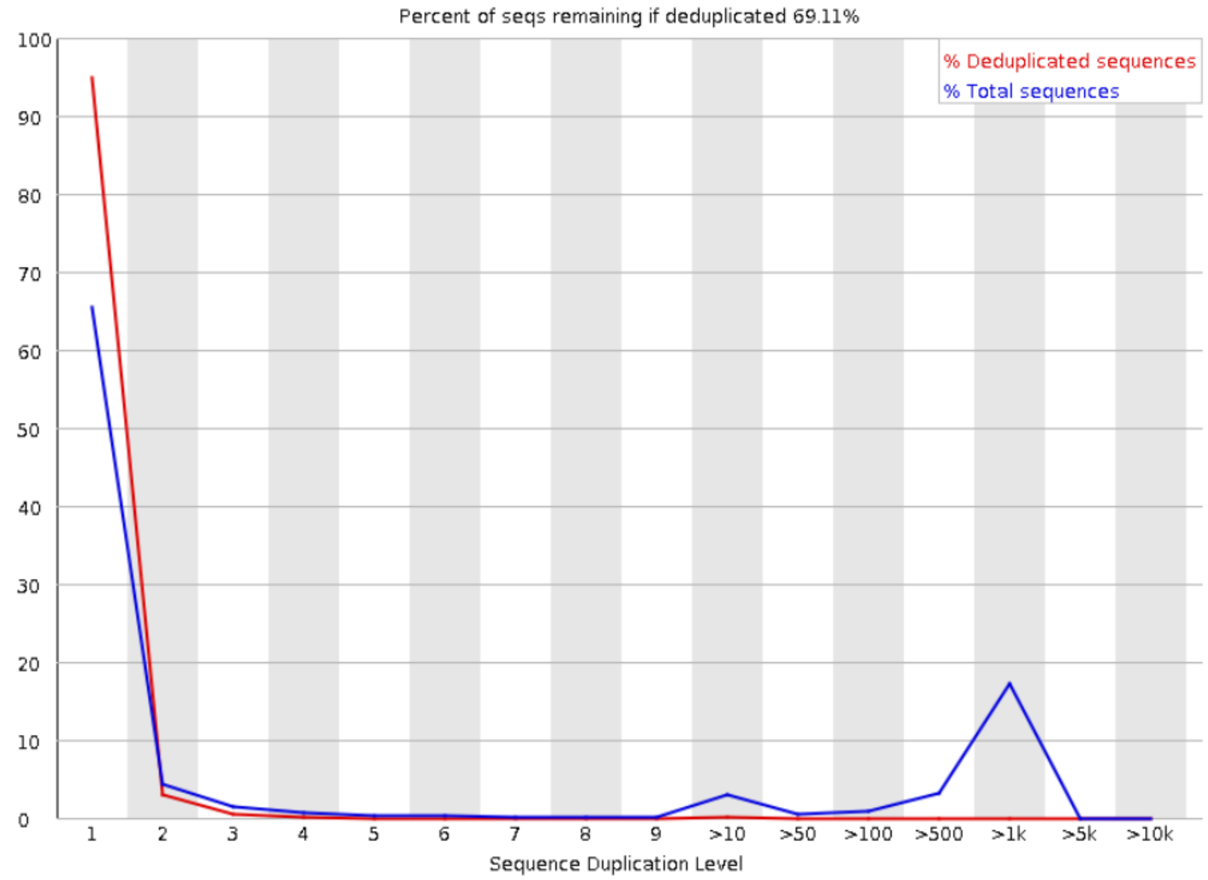
How can I Improve?:
In genomics, two peaks might indicate contamination with another organism.
In transcriptomics, this value varies with the composition of the transcriptome

Read duplication

Good Quality Run



Bad Quality Run



How can I Improve?:
I don’t do much at this point, I assess read duplication after aligment

Overrepresented sequences (more than 0,1% of the total)

Good Quality Run

Bad Quality Run

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
AGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACTTTC	2065	0.5224039181558763	No Hit
GATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCATG	2047	0.5178502762542754	No Hit
ATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCATGA	2014	0.5095019327680071	No Hit
CGATAAAAATGATTGGCGTATCCAACCTGCAGAGTTTTAT	1913	0.4839509420979134	No Hit
GTATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAGA	1879	0.47534961850600066	No Hit
AAAAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCT	1846	0.4670012750197325	No Hit
TGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCAT	1841	0.46573637449150995	No Hit
AACCTGCAGAGTTTTATCGCTTCCATGACGCAGAAGTTAA	1836	0.46447147396328753	No Hit
GATAAAAATGATTGGCGTATCCAACCTGCAGAGTTTTATC	1831	0.4632065734350651	No Hit
AAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTC	1779	0.45005160794155147	No Hit
ATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCA	1779	0.45005160794155147	No Hit

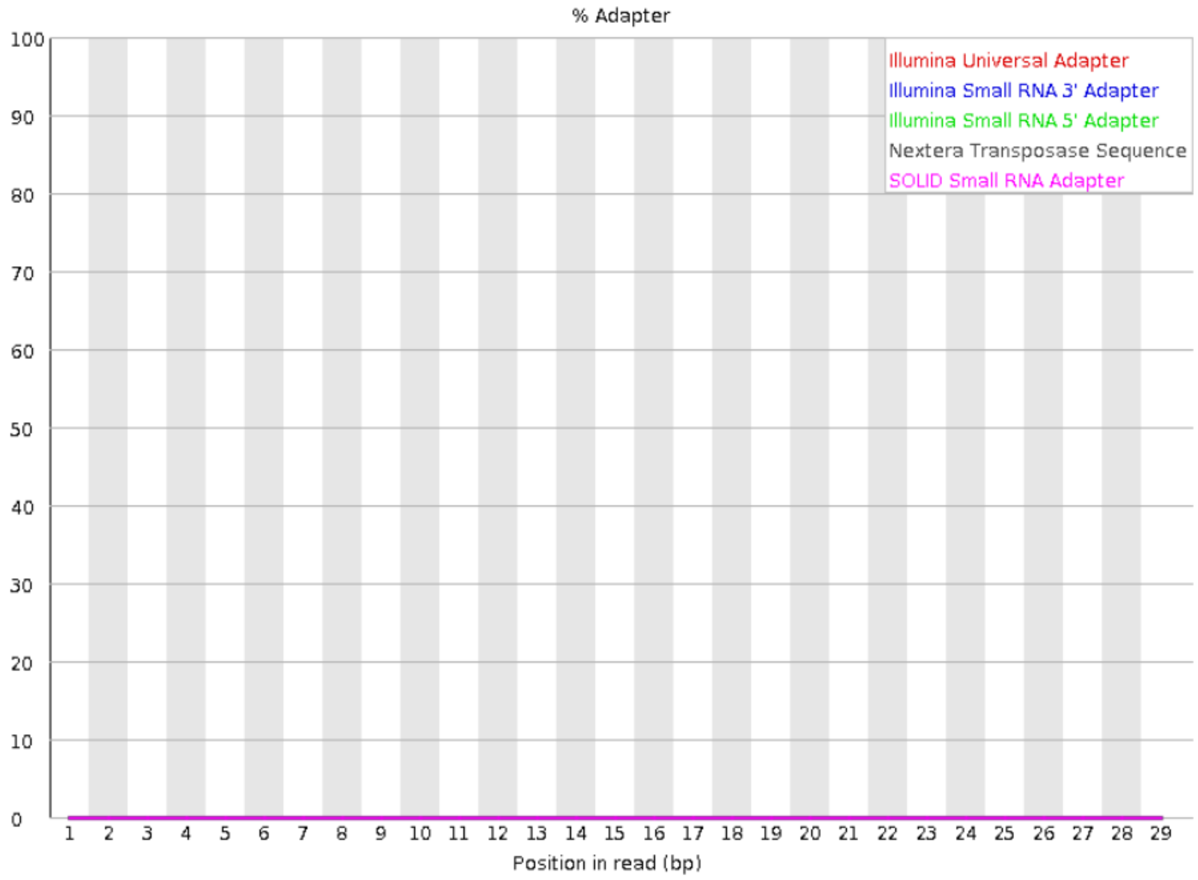
How can I Improve?:

RNA-seq will tend to have overrepresented sequences coming from highly expressed genes

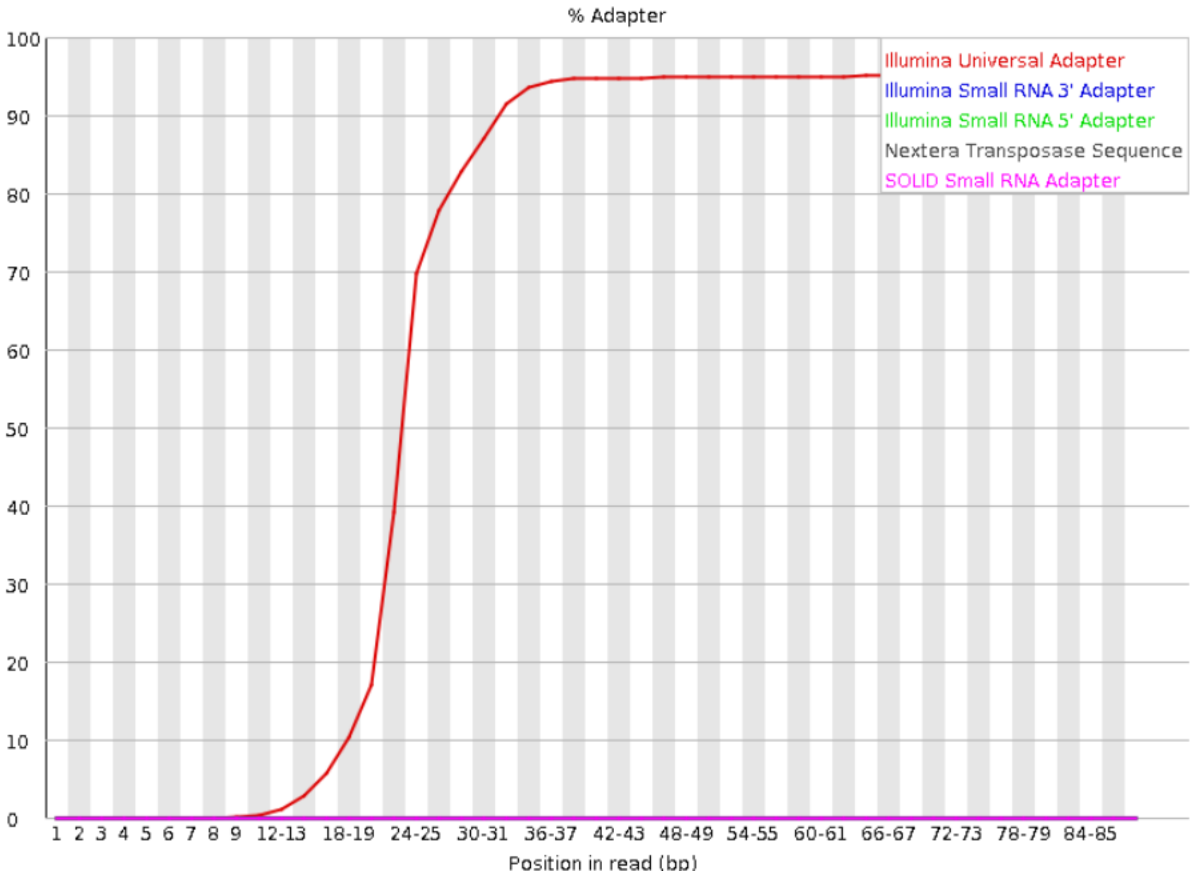
Just blast the sequences listed on the report to be such makes sense with your sample, otherwise it might be contamination

Adapter Content

Good Quality Run



Bad Quality Run



How can I Improve?:

Clip adapter from your sequences using the pre-processing programs :D